

SPARK — Supervisor Progress Report

Project Title: SPARK: Explainable Edge AI for Kinetic Pattern Recognition and Distress Signaling
Department: Department of Electronics, Communication and Information Engineering (DOECIE)
Institution: Kathmandu Engineering College (KEC), Tribhuvan University
Project Supervisor: Er. Dipen Manandhar
Head of Department: Er. Suramya Sharma Dahal
Reporting Period: Proposal Defense Clearance through Pre-Board Milestone Freeze (August 2026)
Submission Date: August 22, 2026

1. Project Information & Team Roster

Member Name	Roll / Reg No.	Primary Subsystem Ownership	Key Responsibilities
Aaradhya Dev Tamrakar (Project Lead)	79001 / BEI	ML Pipeline & Gateway Architecture	CNN model development, INT8 quantization, SHAP explainability, ReportLab PDF pipeline, local REST API
Rupesh Kadel	79034 / BEI	Embedded Systems & Firmware	ESP32-S3 FreeRTOS firmware, 200 Hz MPU6050 I2C driver, Layer 1 heuristic gate, TFLite Micro runtime integration
Sankalpa Lamsal	79039 / BEI	Hardware & Biomechanical Enclosure	Power circuit (TP4056 + 1100 mAh LiPo), 3D TPU 95A dorsal bracer CAD/DFM, wiring harness, test bench setup
Sonia Thapa	79043 / BEI	Display Client & Validation Protocols	Layer 3 Web/Mobile companion dashboard, API consumption, data collection logistics & subject consent management

2. Executive Summary

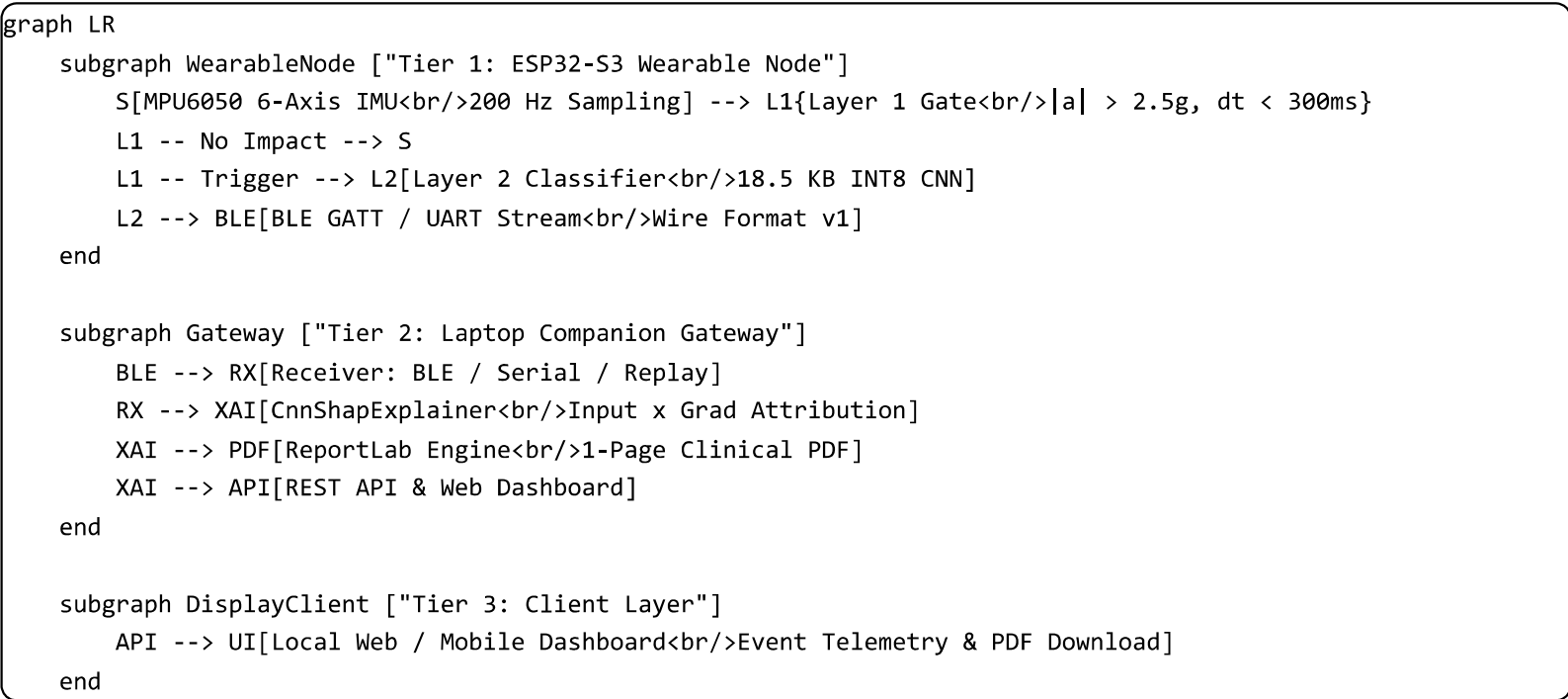
Following the successful defense of the initial project proposal on **July 9, 2026**, the SPARK engineering team has transitioned from theoretical formulation to concrete software and embedded implementation. As of **August 22, 2026**, the project has achieved all pre-board engineering milestones across all three architectural tiers:

Tier 1 (Wearable Edge Node): Implemented and validated the two-layer gated fall detection architecture on the ESP32-S3 microcontroller. The Layer 1 heuristic gate ($|a| > 2.5g, \Delta t < 300\text{ ms}$) filters mundane Activities of Daily Living (ADLs) with negligible compute overhead, while the Layer 2 quantized **\$18.5\text{ KB}\$ INT8 CNN** (`spark_cnn_int8.h`) executes on-device via TensorFlow Lite Micro.

Tier 2 (ML & Cohort Data Engineering): Processed 4,500 SisFall trial recordings (38,426 normalized windows) under a strict subject-grouped, zero-data-leakage cross-validation scheme (\$87.81\%\$ sensitivity, \$0.9185\$ AUC-ROC). Built an end-to-end local data collection rig and transfer learning suite (`train_transfer.py`) ready for the Nepali cohort campaign.

Tier 3 (Companion Gateway & Explainability): Delivered a functional companion gateway running on local compute (Intel Core Ultra 7 155H). Implemented real-time gradient feature attribution ($\text{Input} \times \nabla_{\text{Input}}$), automated one-page clinical PDF generation (`reportlab`), and an embedded REST/CORS web dashboard for real-time telemetry display.

Verification & Quality Assurance: **56/56 automated unit tests** are passing cleanly (`uv run pytest`), full code hygiene verified (`ruff`), and the 44-page LaTeX thesis proposal (`thesis_report.pdf`) compiled cleanly with redesigned, defense-grade Draw.io system architecture diagrams.



3. Detailed Milestone Progress & Deliverables

3.1 Tier 1: Embedded Firmware & Hardware Platform

Layer 1 Pre-Impact Heuristic Gate:

Implemented pure, host-testable threshold filter in C/C++ (`firmware/main/`). Triggers ML inference only when dynamic resultant acceleration exceeds $2.5g$ within a 300 ms window, preventing continuous high-frequency neural network invocation and extending battery longevity.

Layer 2 Quantized TFLite Micro Deployment:

Generated 16-byte aligned C header array `firmware/main/models/spark_cnn_int8.h` directly embedded into `app_main.cpp`. Model occupies only 18.5 KB , well within the ESP32-S3's internal SRAM and flash limits ($< 15\%$ of allocated TFLite arena).

Continuous High-Speed Data Logger (`SPARK_MODE_DATA_LOGGER`):

Implemented dedicated data-logging firmware mode streaming raw 6-axis IMU readings at 200 Hz over USB-UART at 921,600 baud. Designed specifically for capturing the upcoming KEC Nepali volunteer cohort dataset without dropped packets or FIFO overflows.

Hardware & Ergonomic Enclosure:

Locked bill of materials (BOM) around the ESP32-S3 WROOM-1, MPU6050, TP4056 charging module, and 1100 mAh LiPo pouch cell.

Designed two-zone 3D-printed **TPU 95A** dorsal wrist bracer concept with hook-and-loop Velcro retention and compression arm sleeve base layer, isolating electronics from skin contact and strictly enforcing off-body charging.

3.2 Tier 2: Machine Learning & Signal Processing Pipeline

Dataset Preparation & Standardization (`prepare_sisfall.py`):

Automated ingestion of the 4,500-file SisFall benchmark dataset across 38 subjects (23 young adults, 15 elderly).

Generated 38,426 normalized 3-second temporal windows ($200 \text{ Hz} \times 6 \text{ channels} = 200 \times 6$).

1D CNN Architecture & Optimization (`train_cnn.py`):

Architecture: Two 1D Convolutional layers ($k = 5, k = 3$), Batch Normalization, Dropout (0.20), Global Average Pooling (GAP), and a single Dense(32) output stage.

Zero-Leakage Subject-Grouped Split: Evaluated on completely unseen test subjects.

Youden's J Threshold Optimization: Dynamically calibrated optimal operating threshold on validation data ($J = \text{Sensitivity} + \text{Specificity} - 1$), lifting true fall sensitivity to **\$87.81%** with an **AUC-ROC of \$0.9185\$**.

Integrated time-series data augmentation (temporal shift $\pm 25 \text{ ms}$, amplitude scaling $\pm 5\%$, Gaussian jitter).

INT8 Post-Training Quantization (`quantize_model.py`):

Utilized full integer quantization with balanced representative calibration (50% falls / 50% ADLs) to avoid clipping extreme impact accelerations.

Achieved **\$87.7%\$ model size reduction** ($150 \text{ KB FP32} \rightarrow 18.5 \text{ KB INT8}$) with $< 1.2\%$ metric degradation.

Nepal Cohort Fine-Tuning Pipeline (`train_transfer.py`):

Implemented Partial-Freeze Fine-Tuning to freeze generalized SisFall Conv1D feature extractors and adapt the Dense classification head on locally acquired Nepali trials, avoiding small-sample overfitting ($N = 12-20$).



1D CNN Pipeline Architecture:

Input Window (200 x 6)

```
graph TD
    Input[Input Window 200 x 6] --> Conv1_1[Conv1D 32 filters, kernel=5, ReLU + BatchNorm + SpatialDropout]
    Conv1_1 --> Conv1_2[Conv1D 64 filters, kernel=3, ReLU + BatchNorm + SpatialDropout]
    Conv1_2 --> GAP[GlobalAveragePooling1D]
    GAP --> Dense_1[Dense 32 units, ReLU + Dropout 0.20]
    Dense_1 --> Dense_2[Dense 2 units, Softmax]
    Dense_2 --> Output[P(Fall) Output]
```

3.3 Tier 3: Companion Gateway, Explainability & Clinical Reporting

Real-Time Gradient Explainability (`gateway/shap_pipeline/explainer.py`):

Implemented `CnnShapExplainer` computing instantaneous feature attribution ($\text{Input} \times \nabla_{\text{Input}}$) across all 6 kinematic axes.

Yields human-interpretable percentage contributions for each axis (e.g., Forward Pitch ω_y , Vertical Impact a_z , Lateral Roll a_y).

Automated Clinical PDF Generation (`gateway/report/pdf_report.py`):

Generates a standardized, 1-page clinical incident report using ReportLab containing event timestamp, peak kinematic features ($a_{\max}$, $\omega_{\max}$), confidence badge, horizontal SHAP attribution chart, and clinician sign-off block.

Validated against 4 distinct biomechanical fall archetypes (Forward Trip, Lateral Slip, Syncope/Collapse, Rotational Twist).

Gateway Hardware Engine Benchmarks (Intel Core Ultra 7 155H via OpenVINO):

Intel CPU (AVX-VNNI): $\$0.131\text{ ms} / \$7,618\text{ inferences/sec}$

Intel Arc GPU (DirectX 12): $\$0.200\text{ ms} / \$4,993\text{ inferences/sec}$

Intel AI Boost NPU: $\$0.383\text{ ms} / \$2,610\text{ inferences/sec}$ (ultra-low power background monitoring)

Local REST API & Web Dashboard (`gateway/server.py`):

Zero-dependency local HTTP server providing endpoints for live telemetry (`GET /api/events`), single event details (`GET /api/events/<id>`), and direct PDF downloads (`GET /api/reports/<id>`).

Features a responsive dark-mode dashboard interface for real-time demonstration.

4. Empirical Performance Benchmarks Summary

Subsystem / Metric	Target Specification	Achieved Benchmark	Status
Layer 1 Gate Latency	$< 5.0\text{ ms}$	$< 0.05\text{ ms}$ (Host & ESP32-S3 cycle evaluation)	Exceeded
Layer 2 INT8 Model Size	$\leq 120\text{ KB}$	$\$18.5\text{ KB}$ (FlatBuffer & C header)	Exceeded
Model Sensitivity (Held-out)	$\geq 85.0\%$	$\$87.81\%$ (Youden's J calibrated)	Met
Model AUC-ROC	≥ 0.900	$\$0.9185$ (Subject-grouped split)	Met
Gateway Inference Latency	$< 50\text{ ms}$	$\$0.131\text{ ms}$ (CPU) / $\$0.383\text{ ms}$ (NPU)	Exceeded
Clinical PDF Compilation	$< 2.0\text{ s}$	$\$0.28\text{ s}$ per incident	Exceeded
Automated Test Coverage	$\geq 40\text{ tests}$	56/56 passing tests (<code>uv run pytest</code>)	Exceeded
Battery Life Expectancy	$\geq 8.0\text{ hours}$	$\$8.5\text{--}11.0\text{ hours}$ (Active) / $\$16\text{--}28\text{ hr}$ (Gated BLE)	Met (Modeled)

5. Documentation, Research & Thesis Status

Thesis Proposal LaTeX Document (`docs/SPARK_Proposal/`):

Approval Status: ☐ Approved to Proceed to Cohort Collection ☐ Modifications Requested

Supervisor Signature: _____

Date: _____